

A Neuron-Inspired 0.0032mm²-1.38μW/Ch Wireless Implantable Neural Interface with Direct Multiplexing Front-End and Event-Driven Spike Detection and Transmission

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The channel count of implantable neural interfaces has grown rapidly as brain-machine interface applications emerge from basic brain disorder monitoring, motor control, to voice reconstruction. However, limited on-implant power and area budget hinder the development of neural interfaces and multi-channel integration. As shown in Fig. 1, Although time-division multiplexing (TDM) before ADC [1,2] alleviates the area consumption over architecture without TDM [3], the recording front-end still occupies large on-chip area, resulting average channel area of over 0.01mm². Recently, TDM over the entire readout circuits has been adopted to reduce area consumption per channel [4,5]. However, they induce extra overhead to resolve DC offset variations between channels. More importantly, the power consumption issue associated with large volumes of neural signal transmission remains unsolved.

Inspired by the energy and area efficient neural data interaction of the biological neurons, this paper proposes a neuromorphic neural interface (NNI) design shown in Fig. 2 to solve the aforementioned challenges. Firstly, the high fan-in nature of biological neurons is realized in the NNI by multiplexing the entire readout circuit. A low area budget of 0.0032 mm² per channel is achieved even with the dedicated electrode DC offset (EDO) cancellation method. After multi-channel neural recording, the signal sampling, processing and transmission of the NNI all obey the neuromorphic principle where information is represented and processed as spikes with event-driven characteristics. Specifically, a digital spike train representing the recorded neural information is generated with an analog-to-spike converter (ASC). It realizes adaptive sampling based on neuron firing rates to avoid unnecessary sampling power consumption. A successive clockless spike detector is triggered by the digital spike output of ASC in an event-based manner to further reduce clocking power, with which the raw neural activities are compressed and represented as digital spikes, achieving more than 500x data reduction. Finally, the highly compressed digital spikes wake up the transmitter for data transmission. With the proposed neuromorphic approach, NNI can effectively deal with a large volume of neural signals, while consuming only 1.38μW per recording channel.

Fig. 3 depicts the proposed 32-channel neural interface. Addressing fluctuated offsets from electrochemical polarization at the electrode-tissue interface is a challenge in direct TDM-based front-end designs. To mitigate the risk of amplifier saturation, a digital DC servo loop is proposed to leverage the offset-insensitive ASC advantage. During recording, the neural interface first enters the offset calibration mode, where the second stage transconductor switches to a small gm setting and all sampling switches are closed. The parallel connection of sampling capacitors forms a large capacitor (32x) along with the small gm (/100), creating an integrator with a low cut-off frequency. Following the reset signal, the ASC quantifies the offset drift originating from VCM. The offset is then reconstructed as a 6b code representing the average offset from the 32 channels using a dual-counter based on the up and down spikes generated by the ASC, and fed back to the input through a 6b capacitive DAC. The digital DC servo loop effectively prevents amplifier saturation while maintaining efficiency in area and power consumption. This calibration process has a settling time of 0.96ms and is executed every 4.8s to accommodate the slow change characteristic of the offset. After the calibration phase, the neural interface monitors neural signals using a closed-loop programmable-gain LNA with a current-reuse amplifier. Additionally, a Gm-C integrator provides approximately 20dB of extra gain and enables anti-aliasing. To minimize crosstalk between adjacent channels, periodic reset operations are performed on the intermediate nodes, and the KT/C noise is up-modulated by choppers and subsequently filtered out. A

demux is used to separate multiplexed neural signals for conversion to digital spikes and neural spike detection. The subsequent 32-channel ASC and spike detector are working parallelly with an event-driven mechanism.

Fig. 3 (bottom) and Fig. 4 (top) depict the ASC's architecture and circuit schematics. It consists of a 1-bit capacitive DAC-based signal folding circuit, two asynchronous comparators, and a biologically-inspired refractory circuit. The refractory circuit generates outputs (UP/DN and CLK_UP/CLK_DN) based on predefined thresholds (VH and VL). The UP/DN outputs connect to the subsequent spike detector, while CLK_UP/CLK_DN act as refractory periods and control signals for the signal folding circuit and asynchronous comparators. The signal folding circuit resets input signals to VM upon receiving CLK_UP/CLK_DN, and the refractory period determines the ASC's maximum operating frequency and output data rate. The 3-stage asynchronous comparators are used. To reduce static power consumption, power-gating PMOS transistors are incorporated and controlled by the refractory circuit, effectively turning off the comparators during signal folding. The refractory circuit includes a PMOS common source stage connected with inverter buffers and an adjustable NMOS load. By adjusting the refractory voltage of the NMOS load, the circuit generates asynchronous clock signals with a variable refractory period, enabling reconfigurable sampling and adaptability to different neuron firing rates. Compared to previous works using Nyquist sampling ADC, the proposed ASC achieves over 100x data reduction through sampled data reduction techniques.

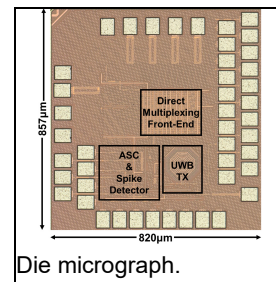
Detecting neural spikes is crucial for reduced data transmission and power consumption. The proposed event-driven spike detector, depicted in Fig. 3 bottom, utilizes a valley-to-peak relative threshold inspired by the neuron integrate-and-fire principle. This approach enhances the detector's tolerance to noise. When receiving UP and DN outputs from the ASC, the spike detector engages in event-driven detection utilizing an inner valley/peak detector, counter, and comparator, ultimately outputting the detected spikes. Moreover, one spike detector occupies a small area of 0.0007mm² and low power of 0.48μW, enabling multi-channel integration.

The proposed neural interface integrates a serializer data interface and an IR-UWB transmitter, which achieves event-driven spike transmission by selectively activating the data transmission module upon detecting action potentials. As depicted in Fig. 4 top right, the IR-UWB transmitter consists of three parts: impulse generator, oscillator, and digital power amplifier. When the detected neural spike arrives, the impulse generator will produce enable signals to wake up the oscillator and the digital power amplifier for 1-bit data transmission. Given the condition of simultaneous 32-channel neuron firing rate transmission, the transmitter only consumes 4.6μW.

Fig. 4 bottom demonstrates the proposed 32-channel neural interface operation with in vitro rat neural signal generator, showing multiplexed neural recording, digital spike based event-driven signal sampling and data reduction, and the transmitter transient output waveform when a neural spike is detected. Fig. 5 presents the AFE gain programmability, dynamic performance, input-referred noise, the ASC SNDR, and the TX output spectrum under 25Mbps input data rate, obeying FCC regulation. Fig. 6 shows the comparison to the state of the art works. The 32-channel neural interface occupies 0.0032 mm² -1.38μW per recording channel. Through neuron-inspired system, module, and circuit level optimizations, the proposed neural interface outperforms the state of the art in per recording channel power, and more than 500x data reduction for low power transmission.

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References: [1] D. Yoon *et al.*, VLSI 2021. [2] Y. Jia *et al.*, ISSCC 2020. [3] R. Muller *et al.*, JSSC 2015. [4] X. Huang *et al.*, JSSC 2022. [5] U. Shin *et al.*, ISSCC 2022. [6] C. Yang *et al.*, ISSCC 2023.



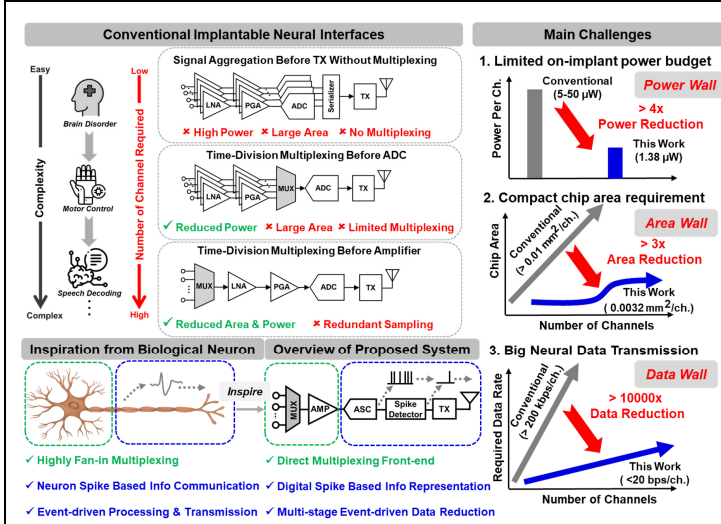


Fig. 1. Conventional implantable neural interfaces (top) with main challenges (top, right), and overview of proposed system inspired from biological neuron (bottom).

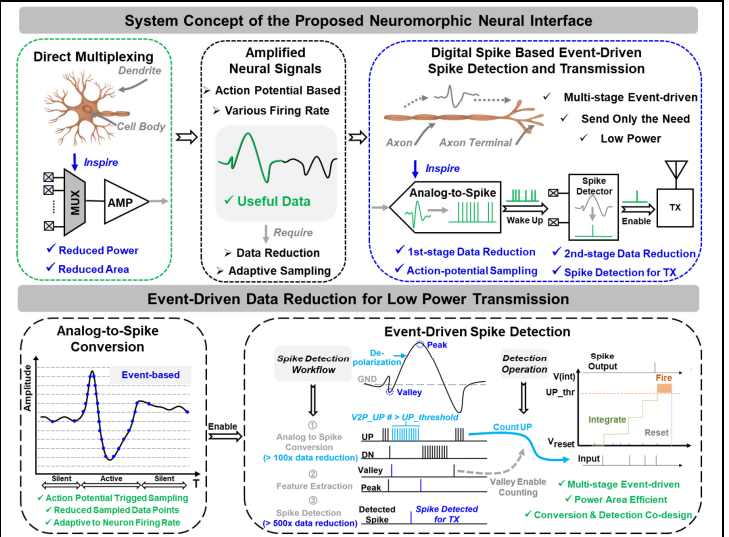


Fig. 2. System concept of the proposed neuromorphic neural interface (top), including event-driven data reduction for low power transmission (bottom).

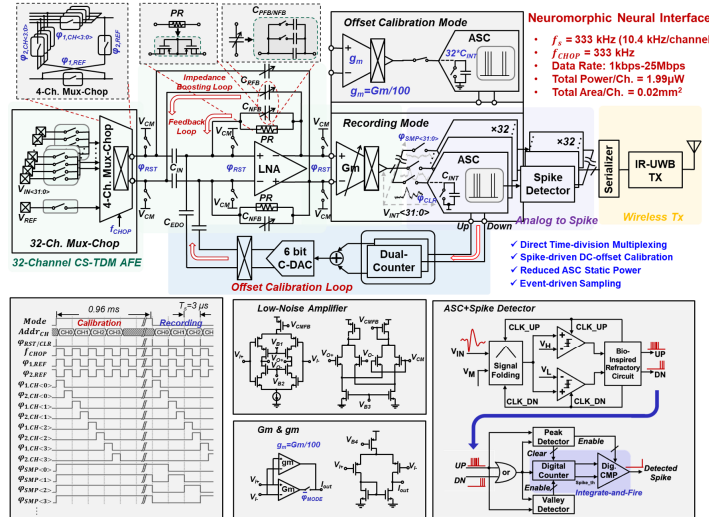


Fig. 3. Detailed system diagrams of the proposed neuromorphic neural interface.

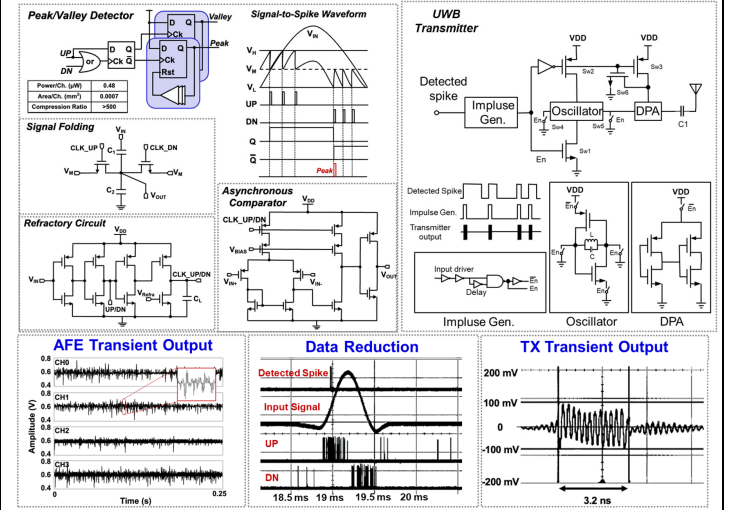


Fig. 4. Schematics of ASC, spike detector, and UWB TX; measured waveform of AFE transient output, data reduction, and TX transient output.

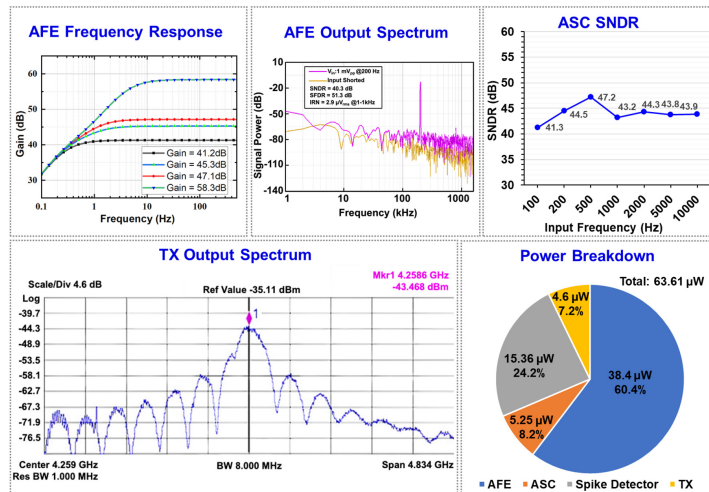


Fig. 5. Measurement results: AFE performance, ASC performance, TX output spectrum under maximum data rate, and power breakdown.

	This work	ISSCC'23 Yang [6]	JSSC'22 Huang [4]	ISSCC'22 Shin [5]	VLSI'21 Yoon [1]	ISSCC'20 Jia [2]	JSSC'15 Muller [3]
System	Technology (nm)	40	65	22	65	350	65
	Supply Voltage (V)	1.2/0.6	-	0.8	1.2	2.5	1.8/3.3/4
	# Recording Ch.	32	128	256	256	1024	16
	Total Power (μ W)	63.61	1098	412.16	453	24658	2270
	Chip Area (mm ²)	0.7	4	0.256	3.48	20.48	15
	Total Power/Ch. (μ W)	1.99	17.2	1.61	1.77	24.08	1294
	Total Area/Ch. (mm ²)	0.02	0.031	0.001	0.014	0.02	0.94
Neural Recording	Architecture	TDM + ASC ^a	SC-DDC	I- $\Delta\Sigma$	TDM+SAR	CCIA+SAR	Shared ADC
	Power/Ch. (μ W)	1.38	9.87	1.61	1.51	2.72	50 ^b
	Area/Ch. (mm ²)	0.0032	0.02	0.001	0.004	0.0062	0.32 ^c
	Bandwidth	5.2kHz	10kHz	500Hz	1kHz	5/300Hz-10kHz	1~100Hz-10kHz
Data TX	Simultaneous Recording BW (kHz) ^d	166.5	640	128	256	10240	160
	Avg. IRNoise (nV/rHz)	91	209	69	143	216	34.6
	Data Reduction	Yes	No	No	No	Yes	No
	Data Rate	Neuron Firing Rate	20.16 Mbps	-	-	6.78 Mbps	1 Mbps
Data TX	TX Power (μ W)	4.6	16.1	-	-	9100	2.4
	Energy Eff. (pJ/bit)	17.16 ^e	0.8	-	-	1342	2.4

^a Analog-to-Spike Converter

^b Only amplifier power included

^c Estimated from die photo

^d Simultaneous recording channels x BW

^e Measured at 25Mbps

Fig. 6. Performance comparison to state-of-the-art works.